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Class : SY.BSC.IT	Batch : 1
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6a:- Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records.

```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

Print top 5 values

```
print(df.head())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Kaif	BSc IT	60	85
1	102	Aditya	BSc CS	75	90
2	103	Aryan	BSc IT	80	92
3	104	Usaid	BSc CS	60	80
4	105	Jasad	BSc IT	70	70

Print last 5 values

```
print(df.tail())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Kaif	BSc IT	60	85
1	102	Aditya	BSc CS	75	90
2	103	Aryan	BSc IT	80	92
3	104	Usaid	BSc CS	60	80
4	105	Jasad	BSc IT	70	70

6b:- Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information

Display rows and cloumns

```
print(df.shape) #number of rows and columns
```

```
(5, 5)
```

Display columns names

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
```

Display Data types

```
print(df.dtypes)
```

```
Roll No      int64  
Name         object  
Department   object  
Marks        int64  
Attendance   int64  
dtype: object
```

Display basic statistical information

```
print(df.describe)
```

```
<bound method NDFrame.describe of
0      101      Kaif      BSc IT      60      85
1      102      Aditya     BSc CS      75      90
2      103      Aryan      BSc IT      80      92
3      104      Usaid      BSc CS      60      80
4      105      Jasad      BSc IT      70      70>
```

6c:- Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60 or employees having salary greater than 40,000.

Display students who have greater than 60 marks

```
result = df[df["Marks"]>60]
print(result)
```

```
Roll No      Name Department  Marks  Attendance
1      102      Aditya     BSc CS      75      90
2      103      Aryan      BSc IT      80      92
4      105      Jasad      BSc IT      70      70
```

6e:- Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks or products by price.

Display students by sorting the marks column in ascending order

```
result = df.sort_values("Marks")
print(result)
```

```
Roll No      Name Department  Marks  Attendance
0      101      Kaif      BSc IT      60      85
3      104      Usaid      BSc CS      60      80
4      105      Jasad      BSc IT      70      70
```

1	102	Aditya	BSc CS	75	90
2	103	Aryan	BSc IT	80	92

6d:- Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column.

Display maximum, minimum, average and total values of numerical columns

```
print("Maximum Marks ",df["Marks"].max())  
print("Minimum Marks ",df["Marks"].min())  
print("Sum Marks ",df["Marks"].sum())  
print("Mean Marks ",df["Marks"].mean())
```

```
Maximum Marks  80  
Minimum Marks  60  
Sum Marks  345  
Mean Marks  69.0
```

